

- Functional Requirements Document (FRD) for Data Platform**
- Introduction:**

The purpose of this document is to outline the functional requirements for the Data Platform project, which aims to integrate customer and order data from various sources into a unified view.

- Functional Requirements:**

1. **Data Ingestion**

- The system shall read CSV data from the specified volume location (`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`) into Delta tables for customer and order data.

2. **Data Quality**

- The system shall remove "Null"/Null and duplicate records from both customer and order tables.

3. **Data Transformation and Loading**

- The system shall create an "ordersummary" table if it does not exist in the specified catalog (`gen\_ai\_poc\_databrickscoe`) and schema (`sdlc\_wizard`).
- The system shall join customer and order data using the "CustId" field and load the data into an SCD type 2 table (`ordersummary`).

4. **SCD Type 2 Implementation**

- The system shall implement SCD type 2 logic to update the "ordersummary" table whenever there is a change in the customer table.

5. **Aggregated Reporting**

- The system shall create a new table "customeraggregatespend" with columns "Name", "TotalAmount", and "Date" in the specified catalog (`gen\_ai\_poc\_databrickscoe`) and schema (`sdlc\_wizard`) if it does not exist.
- The system shall aggregate the "TotalAmount" column from the "ordersummary" table and group by "Name" and "Date" columns.

- Data Flow Diagram:**

```
```json
```

```
{
 "name": "generate_dfd_diagram",
 "arguments": {
 "text": "CSV Data ->> Data Ingestion: Read CSV data into Delta tables; Data Ingestion ->> Data Quality: Remove Null/Null and duplicate records; Data Quality ->> Data Transformation and Loading: Join customer and order data; Data Transformation and Loading ->> ordersummary Table: Load data into SCD type 2 table; ordersummary Table ->> SCD Type 2 Implementation: Update table on customer data changes; ordersummary Table ->> Aggregated Reporting: Aggregate TotalAmount by Name and Date; Aggregated Reporting ->>"
 }
}
```

customeraggregatespend Table: Load aggregated data into table"}}}

...

- System Components:\*\*

1. Data Ingestion Module
2. Data Quality Module
3. Data Transformation and Loading Module
4. SCD Type 2 Implementation Module
5. Aggregated Reporting Module

- Interface Requirements:\*\*

1. The system shall read CSV data from the specified volume location.
2. The system shall write data to the Delta tables, "ordersummary" table, and "customeraggregatespend" table.

- Data Requirements:\*\*

1. The system shall handle large volumes of data.
2. The system shall ensure data accuracy and consistency through data quality checks.

- Success Criteria:\*\*

1. The data is successfully ingested into the Delta tables.
2. The data quality checks are passed, and the data is accurate and consistent.
3. The "ordersummary" table is created and populated with the joined data.
4. The SCD type 2 logic is implemented correctly, and the "ordersummary" table is updated accordingly.
5. The "customeraggregatespend" table is created and populated with the aggregated data.